

Business Understanding Report

Employee Attrition Analysis

AnalystLab Africa – Data Science Internship (Week 1)

Introduction

Employee attrition refers to employees leaving a company due to resignation, retirement, or other reasons. While some employee turnover is expected, a high attrition rate can negatively affect an organization. Companies lose experienced workers, spend more money recruiting and training new employees, and may experience lower productivity. According to the assignment, ABC Manufacturing Ltd wants to understand why employees are leaving before investing in predictive machine learning models.

Business Problem

ABC Manufacturing Ltd has noticed that employees are leaving the company, and the Human Resources (HR) department wants to identify the possible reasons behind this trend. Instead of making assumptions, the company plans to use data to understand employee behaviour and make informed decisions. This analysis will help the company improve employee retention and create a better working environment.

Why Employee Attrition Matters

High employee attrition affects both the company and its employees. When experienced employees leave, the company loses valuable skills and knowledge. Hiring and training new staff also require time and money. Frequent employee turnover can reduce team performance, lower employee morale, and delay business operations. Understanding the causes of attrition allows the company to take action before the problem becomes more serious.

How Data Science Supports HR

Data science helps HR departments make decisions based on facts rather than guesses. By analysing employee data, organizations can identify patterns and trends that may be linked to attrition. For example, data analysis can show whether employees who work overtime, earn lower salaries, or belong to certain departments are more likely to leave. These insights help HR develop better policies, improve employee satisfaction, and reduce staff turnover.

In the future, the same data can also be used to build machine learning models that predict which employees are at risk of leaving, allowing the company to intervene early.

Project Objectives

The main objective of this project is to explore the employee dataset and answer important business questions about employee attrition. The analysis aims to:

Understand the company's workforce.

Identify departments with the highest employee attrition.

Determine whether age affects attrition.

Examine the relationship between monthly income and employee retention.

Investigate whether overtime influences employee attrition.

Identify job roles with the highest turnover.

Discover variables that may be useful for future predictive models.

Expected Business Value

The findings from this analysis will help ABC Manufacturing Ltd better understand its workforce and the possible reasons employees leave the company. These insights can support HR in making better decisions about employee welfare, workload, salaries, and retention strategies. They also provide a strong foundation for future predictive analytics and machine learning projects.

Conclusion

Employee attrition is an important business issue that can affect productivity, costs, and long-term organizational success. By exploring employee data and identifying key factors related to attrition, ABC Manufacturing Ltd can make informed decisions to improve employee retention. This project is the first step toward using data science to solve real business problems and support better HR decision-making.